

- 2 The nature of the genetic code of a gene is important in determining the sequence of amino acids in a polypeptide chain.

The diagrams show the genetic code for some amino acids and the sequence of bases in part of a gene.

Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid	Genetic code	Amino acid
AAA AAG	Lysine	CAA CAG	Glutamine	GAA GAG	Glutamic acid	TAC TAT	Tyrosine
AAC AAT	Asparagine	CAT CAC	Histidine	GAC GAT	Aspartate	TCA TCC TCG TCT	Serine
ACA ACC ACG ACT	Threonine	CCA CCC CCG CCT	Proline	GCA GCC GCG GCT	Alanine	TGG	Tryptophan

Part of a gene:



The genetic code is described as a triplet code that is non-overlapping and is degenerate.

Explain what each of these phrases means.

Use the information provided to illustrate your answer.

(a) Triplet code

(2)



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(b) Non-overlapping code

(2)

(c) Degenerate code

(2)

(Total for Question 2 = 6 marks)